

CAT 2: Poisson Processes

Deadline on 5th September at 11.59PM (You can use either R or PYTHON)

1. Simulating Poisson Arrivals

- a) Simulate arrival times of patients to an emergency department over a 24-hour period, assuming arrivals follow a Poisson process with rate $\lambda = 5$ patients/hour.
- b) **Task:** Plot the cumulative number of arrivals over time and verify whether the interarrival times follow an exponential distribution.

2. Checking Poisson Assumptions

- a) Given hourly arrival data of vehicles at a toll gate for 7 consecutive days, test whether the counts follow a Poisson distribution.
- b) **Task:** Use a histogram, overlay theoretical Poisson probabilities, and perform a Chi-square goodness-of-fit test.

3. Confidence Interval for λ

- a) For a call center, the number of calls per hour for 30 hours is recorded.
- b) **Task:** Estimate λ (average arrival rate) and construct a 95% confidence interval using the sample mean.

4. Stationarity vs Non-Stationarity

- a) A train station records the number of passengers arriving per 10-minute interval over a day.
- b) **Task:** Using a time series plot and rolling averages, assess whether the process is stationary (constant rate) or time-varying.

5. Superposition of Poisson Processes

- a) Two independent Poisson processes model arrival of male and female customers at a service desk ($\lambda_1 = 3/\text{hour}$, $\lambda_2 = 2/\text{hour}$).
- b) **Task:** Simulate and visualize the combined process. Derive the distribution and CI for total arrivals per hour.

6. Modeling Count Data with Poisson Regression

- a. You have **arrival counts per hour** (14/h) at a clinic as the dependent variable and **six other count variables** (e.g., number of staff available, number of walk-ins, number of scheduled appointments, number of emergencies, number of cancellations, and number of transfers) as predictors (Simulate).

b. **Task:**

- i. Perform exploratory data analysis (EDA) with correlation plots and pairwise scatterplots.
- ii. Fit a Poisson regression model and interpret regression coefficients.
- iii. Compute 95% confidence intervals for each coefficient.

7. **Goodness-of-Fit & Visualization**

- a. Using the same dataset in 6, check for overdispersion in the Poisson regression model.

b. **Task:**

- i. Create predicted vs. observed count plots.
- ii. Use deviance and Pearson residuals to assess model fit.
- iii. Discuss whether a Negative Binomial model may be more appropriate.

8. **Poisson Regression with ML Framework (Simulate)**

- a. Simulate a sample where $n=2,500$, 14/h arrival and 12 other exploratory variables that are count. Fit a **regularized Poisson regression (e.g., Lasso or Ridge)** using the 12 count predictors to forecast hourly arrivals.

b. **Task:**

- i. Perform train-test split, fit ML-based Poisson model, and evaluate predictive performance using Mean Deviance and Poisson Log-Likelihood.
- ii. Visualize feature importance (e.g., via SHAP values) and discuss insights on predictor impact.